DAY 21 OF 30

Direct Lake

Fast reports without copying data.
How it works, and where it stops.

Import | DirectQuery | Framing

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THE PROBLEM

Power BI always made you pick a compromise.

1	**Import**	Fast, but data is a copy and goes stale
2	**DirectQuery**	Always fresh, but every click is slow
3	**Big tables**	Refresh takes hours

Speed or freshness. **For years you could not have both.**

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THE SIMPLE VERSION

Reading the book in the library.

Import is photocopying the whole library and carrying it home. DirectQuery is phoning the librarian with every question.

Direct Lake is **walking in and reading the book off the shelf**: no copy, no middleman.

✓ **No import**	Data stays in OneLake
✓ **No query hop**	Columns load straight into memory
✓ **Result**	Near-import speed on current data

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HOW IT WORKS

What happens when you open a report

1. The model points at **Delta tables** in OneLake
2. Only the **columns the visual needs** are loaded
3. They go straight into **memory** for fast analysis
4. **Framing** moves the model to the latest table version

Why it is fast

Delta's Parquet files are already columnar. The engine reads them as they are instead of converting them.

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FOR ENGINEERS

The two flavours

Direct Lake on OneLake

Reads Delta tables directly. Can combine several sources and mix with Import tables. No DirectQuery fallback.

Direct Lake on SQL

Goes through a SQL analytics endpoint. Falls back to DirectQuery when it has to, which is slower.

Guardrails depend on capacity size

Each SKU has limits on rows and files per table. Past them, one flavour falls back and the other fails.

Table health shows up here

Many small files and missing V-Order make Direct Lake slower.

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THE HONEST PART

Direct Lake or Import?

DIRECT LAKE

Data already lives in OneLake as Delta

Tables are large or refreshed often

You want one copy of the data

IMPORT

Small models where import is simple

Sources outside Fabric

Heavy use of calculated columns

> **The catch:** Direct Lake is only as fast as the Delta tables under it. A table with thousands of small files turns a fast mode into a slow one.

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WATCH OUT

Four things to check

1. **Which flavour** your model uses, and whether it can fall back
2. **Guardrails** for your capacity size
3. **V-Order and compaction** on the tables reports read
4. **Security**: SQL row-level rules do not apply to the OneLake flavour

Direct Lake rewards **good table engineering** more than any other mode.

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SAVE THIS

Direct Lake in five lines

1. **No import**, no copy of the data
2. **Reads Delta** files straight into memory
3. **Two flavours**: on OneLake and on SQL
4. **Guardrails** scale with capacity
5. **Healthy tables** make it fast

Tomorrow: Security basics.
Who can see what, and how Fabric decides.

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